BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022 • Paper Code: KT-X-22-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 12 | **Paper Code:** Code C

Note: Q.1 Attempt all questions of section-A by filling the correct option i.e. A, B, C or D on the OMR RESPONSE SHEET. Cutting, erasing and over writing will not be awarded any marks.

Qi. All of these are the methods of vegetative propagation of plants accept.....

(A) cutting
(B) grafting
(C) spore formation
(D) cloning

Qii. A Hydra reproduces asexually by.....

(A) binary fission
(B) budding
(C) spore formation
(D) multiple fission

Qiii. What are the possible genotypes of the parents of homozygous recessive individual (offspring)?

(A) homozygous dominant
(B) homozygous recessive
(C) heterozygous
(D) Both B and C

Qiv. Which one of these bacteria carried out denitrification?

(A) Nitrobacter
(B) Pseudomonas
(C) Rhizobium
(D) Lactobacillus

Qv. One of the following event is not associated with the activity of inspiration.

(A) inter-costal muscles contract
(B) diaphragm becomes more dome shaped
(C) volume of the chest cavity increases
(D) pressure on the lungs decreases

Qvi. Parts of the neuron that carry signals towards the cell body is called.....

(A) dendrite
(B) axon
(C) Schwann cell
(D) node of Ranvier

Qvii. Which one of the following help in maintaining balance of the body?

(A) cochlea
(B) Eustachian tube
(C) organ of corti
(D) semi-circular canals

Qviii. The point of attachment of a muscle to a fixed or immoveable bone is called.....

(A) insertion
(B) origin
(C) antagonism
(D) agonism

Qix. The tube between kidney and urinary bladder is the.....

(A) urethra
(B) renal tubule
(C) ureter
(D) nephron

Qx. Which one of these gases is the source of global warming?

(A) sulphur dioxide
(B) carbon dioxide
(C) nitrogen oxide
(D) chlorofluoro carbons

Qxi. Interferons are chemicals which are used to.....

(A) kill viruses
(B) improve immunity
(C) work as a vaccine
(D) all of these

Qxii. Endonucleases used in genetic engineering are naturally present in.....

(A) bacterial cell
(B) bacteriophages
(C) plasmids
(D) blue green algae

Section "B"

Time Allowed: 2 Hours 40 minutes. | **Maximum Marks:** 32

QQ.2. Attempt any EIGHT parts. All parts carry equal marks.

i Differentiate between the term respiration and breathing.

ii What is the role of insulin and glucagon.

iii Differentiate between transpiration and guttation.

iv How are different types of arthritis caused?

v What is parthenogenesis? Explain it briefly.

vi Define the following terms. a. Genotype b. Phenotype c. Homozygous d. Heterozygous

vii Differentiate between food chain and food web.

viii Differentiate between alcoholic fermentation and lactic acid fermentation.

ix Distinguish between Herbivores, Carnivores and Omnivores giving one example of each.

x What roles do drugs play in our life?

xi Why are bones considered as dynamic structures?

Section "C"

Maximum Marks: 21

Note: Note: Attempt any THREE of the following questions. All questions carry equal marks.

QQ.3.

a. [4] Explain the mechanism of breathing in human.

b. [3] Describe the structure of nephron.

QQ.4.

a. [4] Discuss short sightedness and long sightedness of the human eye.

b. [3] Discuss different types of joints.

QQ.5.

a. [4] Explain Mendel's law of segregation.

b. [3] Differentiate between incomplete dominance and co-dominance.

QQ.6. Write a note on any two of the following.

a. [3.5] Mutualism and commensalism.

b. [3.5] Adaptations for insects and wind pollinated flowers.

c. [3.5] Discuss causes, symptoms and treatment of Epilepsy.